Calibration and Uncertainty of Molecular Property Prediction Models under Distribution Shift: Progress Report

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1 Introduction

The reliability of machine learning models in drug discovery is critical, especially when models are deployed to prioritize candidate compounds for experimental validation. In this project, we investigate the behavior of Graph Neural Networks (GNNs) when subjected to distribution shift, induced by scaffold-based data splitting. We aim to quantify the drop in predictive performance and, more importantly, the degradation of calibration and uncertainty estimation, which are vital for safety-critical applications like bioactivity prediction.

2 Theoretical Background

2.1 Graph Neural Networks in Molecular Property Prediction

Molecular property prediction is fundamentally a problem of learning a mapping from a chemical graph to a scalar property. In this context, a molecule is represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where vertices $\mathcal{V}$ denote atoms and edges $\mathcal{E}$ denote chemical bonds. Traditional QSAR methods relied on hand-engineered descriptors (e.g., Morgan fingerprints), which often fail to capture complex spatial dependencies. Graph Neural Networks (GNNs) overcome this by employing a message-passing framework. In each layer l, the hidden state of a node v, denoted as $h_v^{(l)}$, is updated by aggregating information from its neighbors $\mathcal{N}(v)$:

$$h_v^{(l+1)} = \sigma\left(W^{(l)} \cdot \mathrm{AGG}\left(\{h_u^{(l)} : u \in \mathcal{N}(v) \cup \{v\}\}\right)\right) \tag{1}$$

where σ is a non-linear activation function, $W^{(l)}$ is a learnable weight matrix, and AGG is an aggregation function (e.g., sum, mean, or max pooling). After L layers, the node representations are aggregated via a readout function into a graph-level embedding, which is then mapped to the target property. This inductive bias allows GNNs to learn structural hierarchies directly from chemical topology.

2.2 Generalization and Scaffold-based Distribution Shift

The generalization ability of GNNs is strictly dependent on the data distribution. A random split assumes that the test set shares the same chemical scaffold distribution as the training set, allowing the model to perform interpolation. However, in real-world drug discovery, models often encounter novel chemical structures. Scaffold splitting groups molecules by their Murcko scaffolds—the foundational cyclic frameworks of a molecule—and ensures that these scaffolds are mutually exclusive across train, validation, and test sets. This introduces a rigorous Out-of-Distribution (OOD) evaluation scenario where the model is forced to extrapolate its knowledge to entirely unseen structural classes. This structural gap reveals whether a model has truly learned underlying chemical principles or merely memorized frequently occurring subgraphs.

2.3 Calibration and Quantifying Uncertainty

Calibration addresses the alignment between a model's predicted probability $\hat{p}$ and its empirical accuracy. A perfectly calibrated model satisfies the condition $P(Y = 1|\hat{P} = p) = p$. However, deep neural networks often suffer from overconfidence, especially under OOD conditions, where they output probabilities close to 0 or 1 despite being incorrect. To quantify this discrepancy, we use the Expected Calibration Error (ECE). The ECE partitions the range of predicted probabilities $[0, 1]$ into M equal-sized bins $\{B_m\}_{m=1}^{M}$. Let n be the total number of predictions, and $acc(B_m)$ and $conf(B_m)$ be the accuracy and average confidence within bin B_m, respectively. The ECE is calculated as the weighted average of the absolute differences:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)| \tag{2}$$

A high ECE indicates that the model's confidence is decoupled from its true performance. In safety-critical QSAR tasks, high ECE implies that the model cannot reliably communicate its epistemic uncertainty, thereby necessitating more robust methods such as deep ensembles or Bayesian approximations to recover the true predictive distribution.

3 Methodology

The baseline model is a Graph Convolutional Network (GCN) with 128 hidden channels and 3 message-passing layers, trained using the Binary Cross-Entropy loss on on the Tox21 dataset, comprising 12 distinct toxicity tasks. Training was conducted for 150 epochs with Early Stopping monitored on the validation set to prevent overfitting. We compare two evaluation scenarios: the standard Random Split and the more rigorous Scaffold-based Split.

4 Results

The final predictive performance metrics are presented in Table 1.

Table 1: Predictive Performance Comparison (Test Set)

Split Strategy	Test ROC-AUC	Test PR-AUC
Random Split	0.7972	0.3380
Scaffold Split	0.7009	0.2904

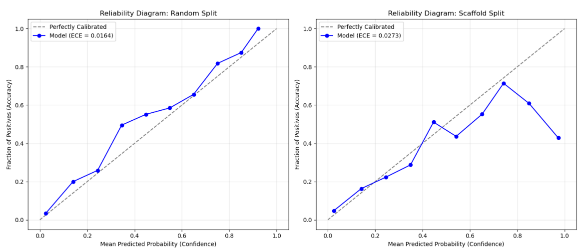

Figure 1: Reliability Diagrams illustrating calibration error (ECE).

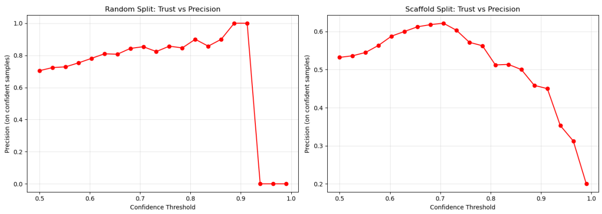

Figure 2: Confidence-Precision curves highlighting model overconfidence.

5 Discussion

The experimental results reveal a significant performance gap and a deterioration of calibration under scaffold-based distribution shift. The drop in ROC-AUC from ≈ 0.80 to ≈ 0.70 provides empirical evidence of the limitations of standard GNNs in generalizing to novel chemical space. This behavior may indicate reliance on frequently occurring local motifs rather than learning more transferable structural principles. Standard GNNs, by design, are excellent at learning local structural features. When training on random splits, these features are shared between training and test sets, allowing the model to perform well via interpolation. However, in scaffold-based splitting, these motifs are effectively "severed" from the test set, forcing the model to extrapolate—a task for which standard GNNs, lacking robust structural priors, are inherently ill-equipped.

The degradation of calibration, evidenced by the increase in ECE and the collapse of the Confidence-Precision curves (Fig. 2), highlights the *overconfidence phenomenon*. This is theoretically driven by the cross-entropy loss function used during training. Cross-entropy encourages the network to maximize the log-likelihood of the correct class, which inherently pushes the sigmoid outputs toward the extremes of $[0, 1]$. While this is beneficial for IID data, it is detrimental under distribution shift. The model becomes "convinced" that its internal feature representations are universal. When it encounters OOD scaffolds, it applies its training-set-derived feature mappings to new nodes, producing high-probability (overconfident) predictions for what are essentially noisy or erroneous inferences.

The inverse correlation observed in the Scaffold-based Confidence-Precision curve (Fig. 2) is particularly alarming for real-world drug discovery. It suggests that the model's confidence is not an indicator of its predictive accuracy, but rather an indicator of its "familiarity" with the input structure. The unexpected degradation of precision at higher confidence thresholds suggests that confidence scores are poorly aligned with predictive correctness under scaffold shift. This behavior classifies the model as a "black box" that fails silently, posing a severe risk in safety-critical molecular screening pipelines where high-confidence false positives lead to significant financial loss in futile experimental validation.

6 Future Work

The established baseline confirms that standard deterministic GNNs lack an explicit mechanism to quantify epistemic uncertainty. While these models can achieve acceptable calibration under i.i.d. conditions (Random Split), their predictive confidence becomes highly unreliable and overconfident under distribution shift (Scaffold Split). This demonstrates that the model's probability output is not a measure of epistemic uncertainty, but rather a reflection of structural similarity to the training data, which collapses when faced with OOD chemical scaffolds. To address this, the project will proceed with the following developments:

- **Deep Ensembles:** We plan to implement deep ensembles, where multiple GNNs are trained with independent random initializations. Deep ensembles are theoretically grounded as a robust method to capture epistemic uncertainty, as the variance in predictions across the ensemble serves as a proxy for the model's uncertainty regarding OOD data.

- **Monte Carlo Dropout:** As a computationally efficient alternative to ensembles, we will implement MC Dropout at inference time. By keeping the Dropout layers active during testing and performing T forward passes, we can approximate the posterior predictive distribution, providing a principled way to estimate uncertainty without the overhead of training multiple models.